Aadil Sood

778-558-7062 | aadilsood014@gmail.com | LinkedIn | Portfolio

Education

Bachelor of Applied Science in Engineering Physics — Vancouver, BC
University of British Columbia — *Sept. 2025 – Present*

Experience

Electrical Team Member — Sept. 2026 – Present
UBC Mars Colony — *Vancouver, BC*

- Developing and integrating serial communication systems across multiple embedded boards using SPI, UART, and I^2C protocols
- Researching and contributing to the design of the Sabatier reactor's mass flow control system, including sensor selection, control hardware, and PCB design for the mass flow controller
- Designing PCBs for the Sabatier project's power distribution system, focusing on reliable power delivery, voltage regulation, and electrical integration across subsystems

Calibration Intern — July 2026 – Aug. 2026
Pylon Electronics – Wescan Calibration — *Richmond, BC*

- Performed precision calibration and verification of dimensional, torque, and electronic measurement instruments
- Developed a Python-based interface to automate data collection from a Laser Scan Micrometer through RS-232 communication, reducing manual measurement entry for technicians
- Operated a wide range of precision metrology, including calipers, micrometers, weighing scales, torque analyzers, and digital multimeters (DMMs)

Projects

STM32-Based Dual Temperature Monitor | *PCB Design, Altium, STM-32* — Sept. 2026 – Oct. 2026

- Designed a custom PCB in Altium featuring an STM32 microcontroller for temperature monitoring within the Sabatier reactor system
- Integrated two MAX6675 thermocouple interface ICs over SPI, alongside voltage regulation, USB-C power, and SWD programming access
- Developed PCB layout and performing design rule checks to evaluate electrical connectivity, component placement, and manufacturability

4-Bit Transistor-Based Binary Adder | *Falstad, BJT Circuits, Digital Logic* — Aug. 2026 – Sept. 2026

- Designed a 4-bit binary adder entirely from discrete BJT transistors, without using integrated logic gate ICs
- Built custom XOR, AND, and other logic gates from transistor networks to implement half and full adder circuits
- Used Falstad Circuit Simulator to prototype, test, and troubleshoot transistor-level logic circuits before physical implementation
- Applied Boolean logic, binary arithmetic, and carry propagation to cascade four 1-bit full adders into a 4-bit arithmetic unit

Laser Scan Micrometer Automation | *Python, Data Automation, Serial Communication* — July 2026 – Aug. 2026

- Developed a Python-based interface that communicates with a Laser Scan Micrometer over a serial connection to automate measurement data collection and streamline the technician workflow
- Troubleshot communication and hardware issues to improve system reliability and ensure consistent operation
- Designed the workflow with future technicians in mind, reducing manual steps and simplifying instrument operation

VEX Robotics Spin Up Robot | *C++, Fusion 360, Assembly Design* — Sept. 2022 – May 2023

- Designed, built, and tested a competitive robotics system to accomplish game objectives efficiently and reliably
- Used CAD and hands-on fabrication techniques to develop mechanical components and assemblies
- Integrated mechanical systems with sensors and control logic to optimize robot performance

Skills

Electrical: PCB Design, Altium, Falstad, Digital Logic, SPI, I^2C, UART, Soldering, ADC/DAC Design, STM-32
Mechanical: Fusion 360, SolidWorks, CAD, Engineering Drawing, Hand and Power Tools, Dimensional Metrology
Software: Python, C, C++, MATLAB, Java, Git, Serial Communication, Data Processing